# Weekly Progress Report

**Name:** Raj Aryan

**Domain:** Machine Learning / Industrial ML

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**Week Ending:** 01

## I. Overview

This week focused on understanding the internship projects, defining the problem statements, inspecting the datasets, and establishing the Python/ML workflow required for the two selected projects: Project 10 (Quality Prediction in a Mining Process) and Project 6 (Remaining Useful Life Prediction for Turbofan Engines).

## II. Achievements

1. Reviewed the objectives, datasets, expected outputs, and industrial relevance of both projects.
2. Project 10: Loaded and inspected the mining-process dataset, identified timestamp/process variables, qualityvariables, and the silica-concentrate target.
3. Project 6: Studied the turbofan engine time-series structure, including unit number, cycle, operating settings, andsensor measurements.
4. Performed initial data-quality checks and identified the need for timestamp handling, missing-value analysis,sensor inspection, and engine-wise sequence preparation.
5. Set up a GitHub-based project structure for reproducible notebooks, source code, documentation, and reports.

## III. Challenges

1. Different sampling frequencies and time scales in the mining-process data require careful temporal alignment.
2. The turbofan data is sequential and engine-specific, so conventional random train/test splitting is notappropriate.
3. Understanding the industrial meaning of process variables and sensor measurements required additional study.

## IV. Learning Resources

1. Internship project document and dataset descriptions.
2. Python, pandas, NumPy, matplotlib, and scikit-learn documentation.
3. Background material on predictive maintenance, RUL prediction, and industrial process quality prediction.

## V. Next Week's Goals

1. Perform detailed EDA and data-quality analysis for both projects.
2. Create the preprocessing pipeline for Project 10, including timestamp handling and resampling.
3. Construct engine-wise sequences and RUL targets for Project 6.
4. Define baseline models and evaluation metrics.

## VI. Additional Comments

The first week established the technical direction for both projects and clarified the difference between conventional tabular ML and time-series/predictive-maintenance workflows.